BSIT102 - Fundamentals of Programming

Semester Project Question:
Algorithm Design and Implementation Using C++

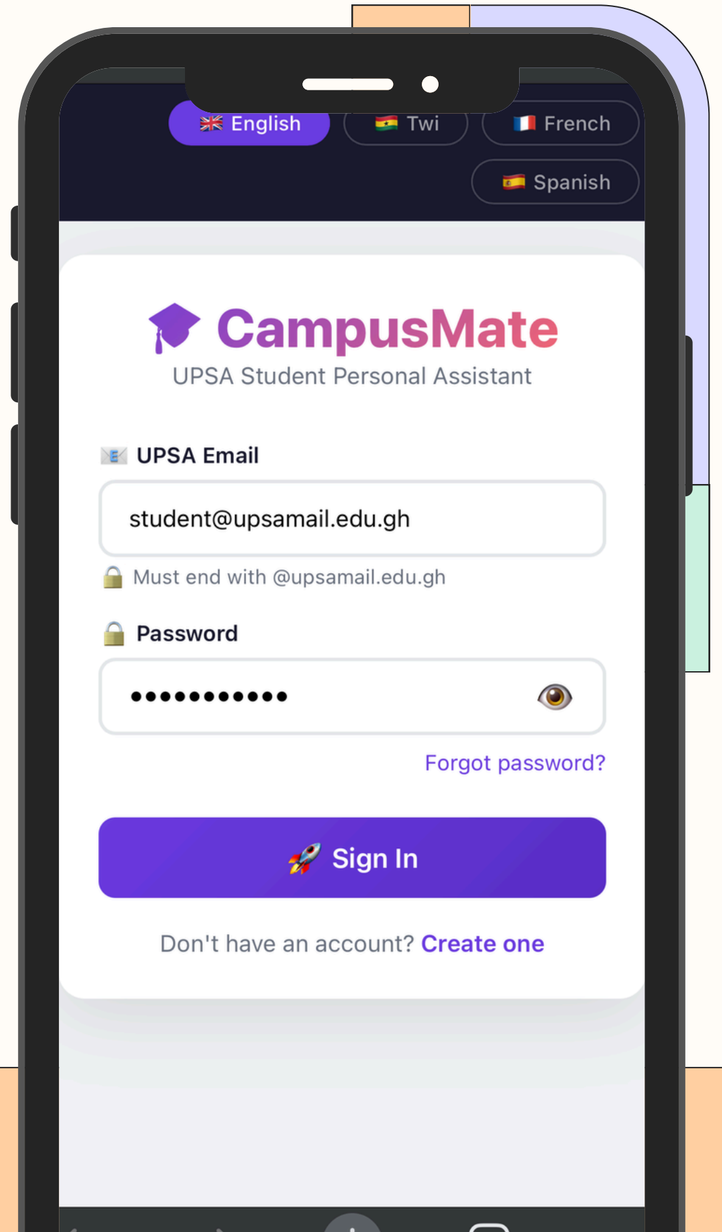
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GROUP 14

CampusMate

A STUDENT ASSISTANT SYSTEM

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• Summary

Your reliable Virtual Student Assistant System

Introduction

What is CampusMate?

- CampusMate is a lightweight, friendly helper program that runs directly on your computer's terminal (the text screen). It acts as a digital planner, keeping all your important university information organized in one private, secure place. It works on different operating systems, made using C++.

Main Idea:

- To connect complex university systems with each student.

Features :

- **Secure Accounts:** Private login and registration to protect your personal details.
- **Course Timetable:** A editable schedule showing where and when your classes are held.
- **Grade Book:** A clean tracker to log your grades and see your CGPA
- **Exam & Test Alerts:** Reminders for IA's and assignments.
- **Class Rep Directory:** Quick access to class rep contact information
- **Notifications:** An bulletin board for announcements.
- **Goal Tracker:** Set academic targets to keep yourself motivated.
- **Automatic Saving:** Keeps your data saved directly on your computer.

- Virtual Learning Site
- Attendance System
- UFIS
- Hostel Booking

Statement of the Problem

The Challenge Students Face

Staying on top of university life is hard when your information is scattered everywhere.

- **Scattered Information:** Class times are on one PDF, grades are on an online portal, and test dates are buried in chaotic group chats.
- **Forgotten Deadlines:** It is incredibly easy to lose track of assignments or miss a sudden mid-semester test.
- **Frustrating Portals:** School websites can be slow, hard to open on weak internet connections, and frustrating to use when you just need a quick update.
- **No Offline Access:** If your internet goes down, you might not even know where your next class is being held.

Objectives of the Project

What We Set Out to Achieve

Our goals for building this software were simple and practical:

1 **One Central Hub**

Combine schedules, grades, and contacts into a single app so you never have to open five different tabs again.

2 **Zero Missed Tests:**

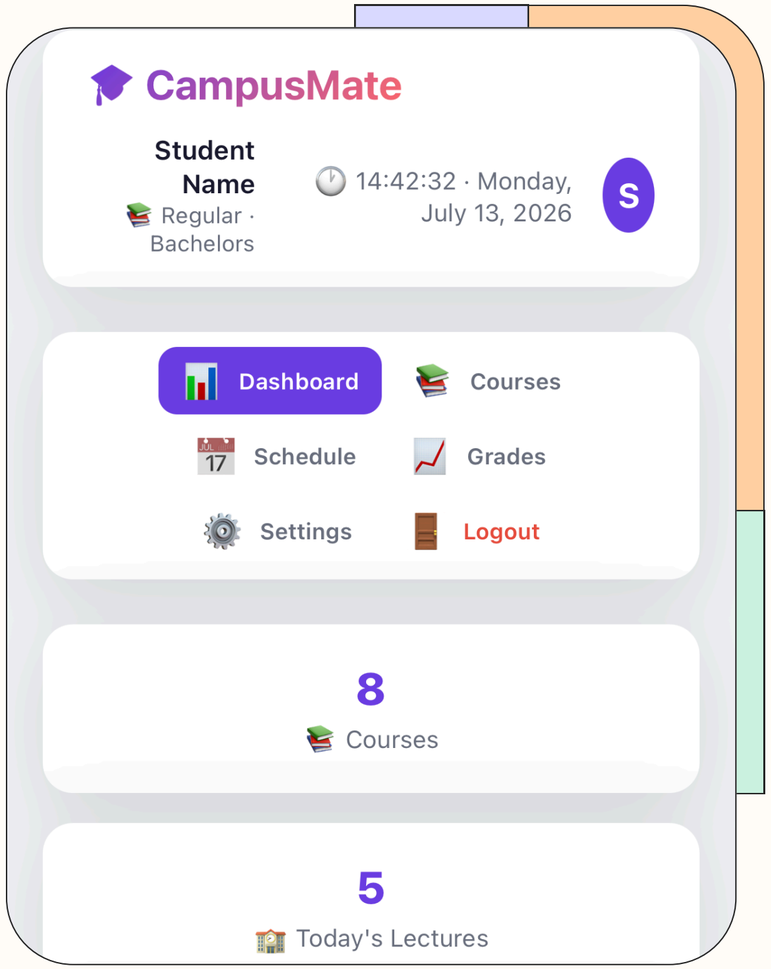
Build a system that automatically flags and reminds you of important IA's as soon as you add them.

3 **Keep You Motivated:**

Let's you set a goal (like First Class), displayed on the dashboard, and helps you see if your displayed CGPA is matching that dream.

4 **No Internet Required:**

Save everything directly to your computer's hard drive so it works perfectly even when you are completely offline.



Scope of the Project

What the Project Covers (and What It Doesn't)

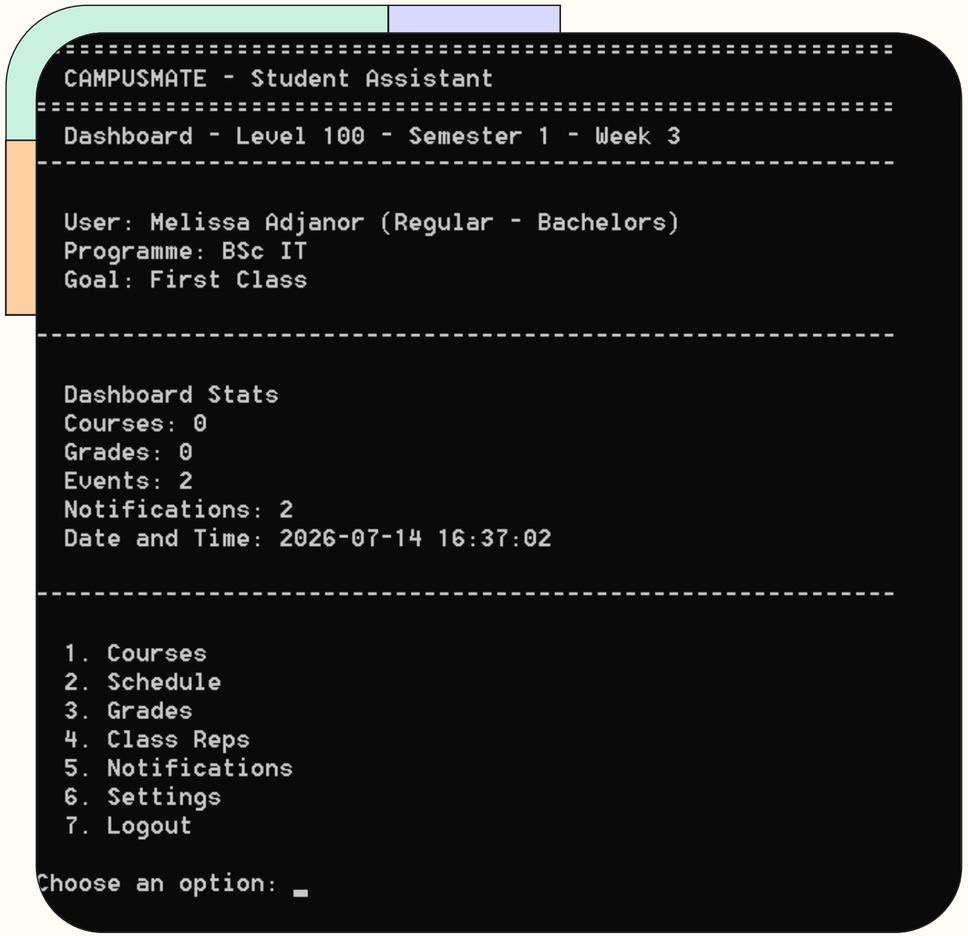
To make this app fast and highly reliable, we set clear boundaries:

- **Who It is For:** Designed for **UPSA students,** using school-specific email checks (requiring a **@upsamail.edu.gh** address) to keep out spam.
- **Where It Runs:** Optimized to run on almost any computer, including Windows, Mac, and Linux.
- **Simple Text Design:** By using a simple text-based screen instead of heavy graphical windows, the app opens instantly and never slows down your computer.
- **Local and Private:** Your information stays on your personal device; nothing is sent to external, risky online servers.

Methodology —
The System Algorithm

Think of the algorithm as the "**recipe**" the computer follows every time you launch **CampusMate:**

- **Step 1:** *Get Ready:* The app boots up, checks your computer for saved files, and loads your previously saved information into its memory.
- **Step 2:** *The Gatekeeper:* It asks you to register or log in. It double-checks that your email is real and matches the school's format.
- **Step 3:** *Load the Dashboard*: Once you log in, it scans your private file, counts your classes, checks for unread notifications, and builds your custom menu.
- **Step 4:** *Wait for Your Command:* It sits patiently and waits for you to choose a menu option, like checking your timetable or adding a new grade.



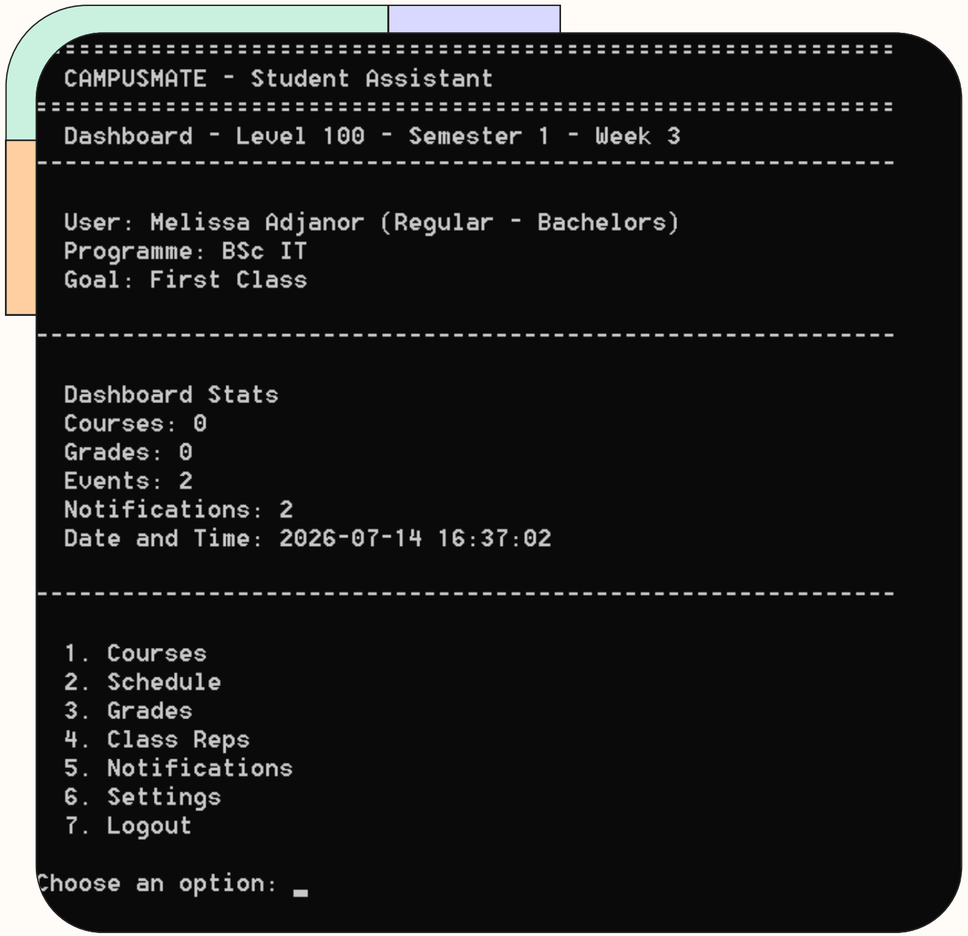
Methodology — *The System Algorithm*

1. **Start**
2. Load all previously saved user and application data.
3. Display the Authentication Menu:
   - Login
   - Register
   - Exit
4. Read the user's choice.
5. If the user selects **Register**:
   - Collect the user's details.
   - Validate the input.
   - Save the new account.
   - Return to the Authentication Menu.
6. Else if the user selects **Login**:
   - Request username and password.
   - Verify the credentials.
   - If the credentials are valid:
     - Set the current user.
     - Display the Dashboard.
   - Otherwise:
     - Display an error message.
     - Return to the Authentication Menu.
7. Else if the user selects **Exit**:
   - Save all data.
   - End the program.
8. While the user is logged in:
   - Display the Dashboard Menu.
   - Read the user's selected option.
9. If **Courses** is selected:
   - Allow the user to add, edit, delete, or view courses.
10. Else if **Schedule** is selected:
    - Display or update the student's timetable.
11. Else if **Grades** is selected:
    - Allow the user to add, edit, view grades, and calculate GPA.
12. Else if **Class Representatives** is selected:
    - Display or manage class representative information.
13. Else if **Notifications** is selected:
    - Display available notifications.
14. Else if **Settings** is selected:
    - Allow the user to modify account or application settings.
15. Else if **Logout** is selected:
    - Save the user's data.
    - Return to the Authentication Menu.
16. Repeat Steps 8–15 until the user logs out or exits the program.
17. Save all application data.
18. Stop

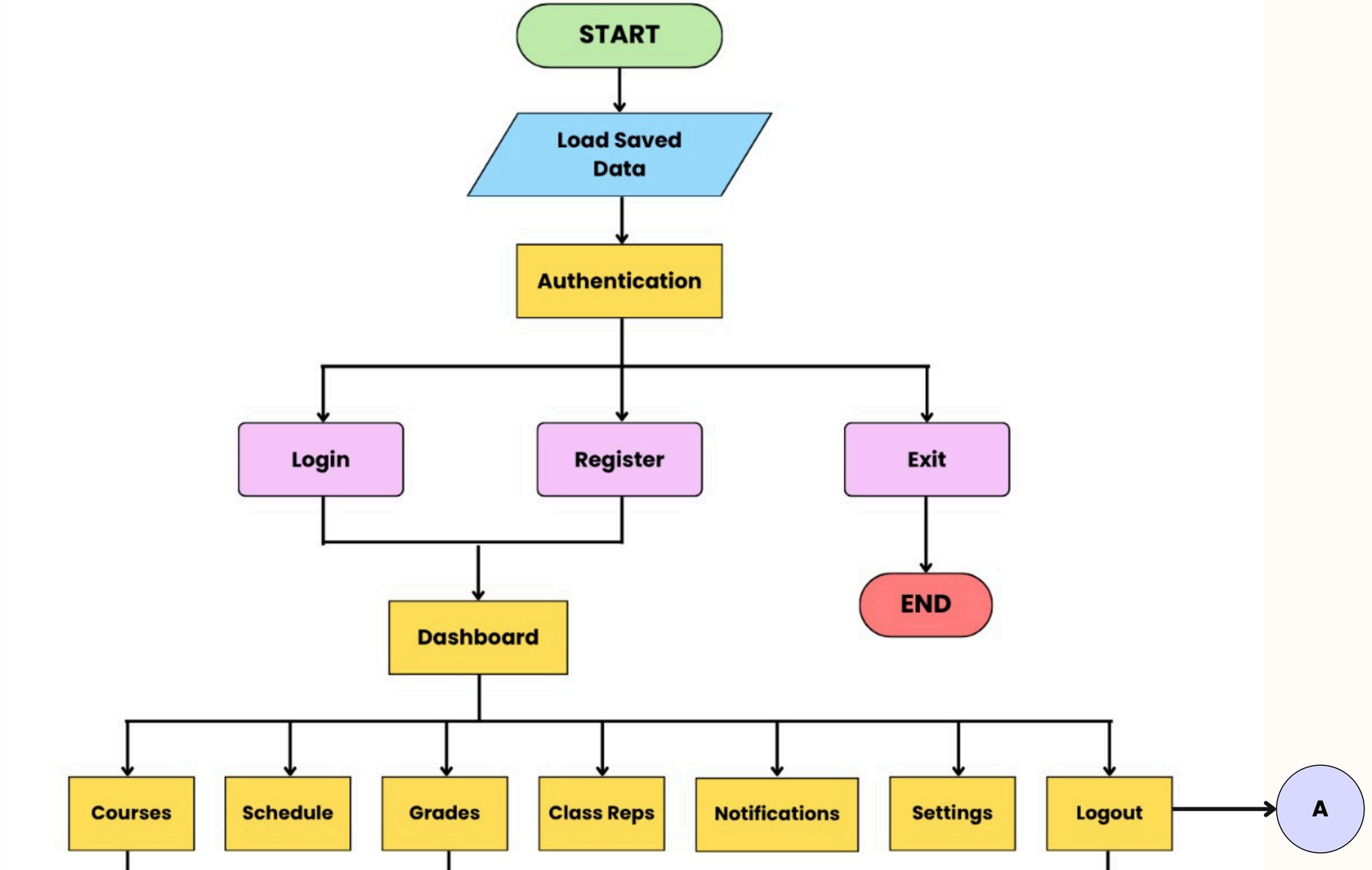
Methodology —
The System Flowchart

The flowchart is a **visual map** showing how a user moves through the app:

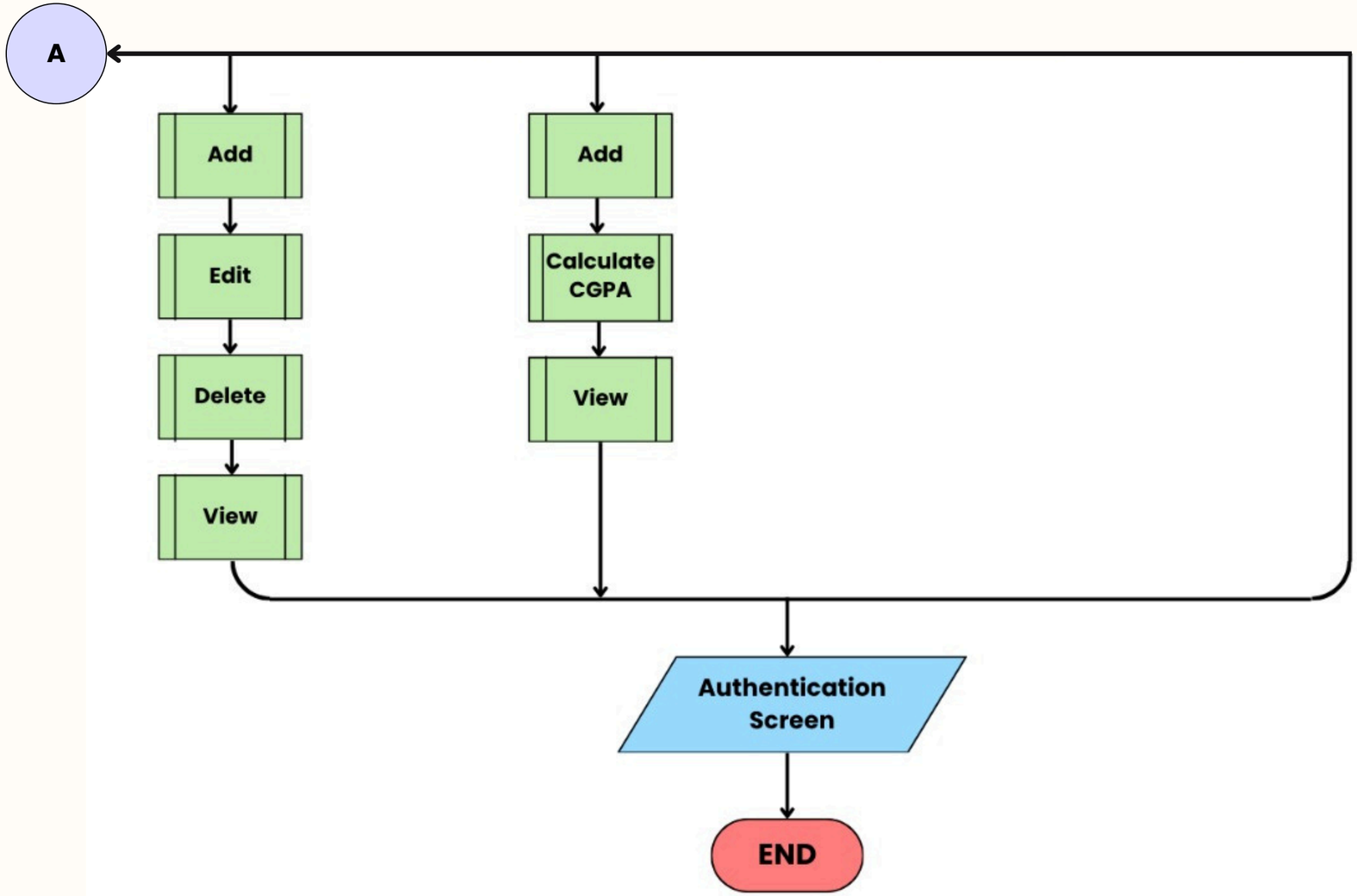
- **The Login :** When you open the app, you stand at a crossroads. You must pass through the security check (**Login or Register)** before you can see anything else.
- **The Main Hub (Dashboard):** Once inside, you are at the central station. From here, you can click on any path you want **(Courses, Grades, Settings etc.).**
- **The Action Loops:** If you choose **"Courses,"** the app takes you to a mini-timetabling screen where you can view, add, or delete classes.
- **The Instant Save:** Whenever you make a change, the app takes a quick detour behind the scenes to write that change to your hard drive before returning you to the main screen.



Methodology — *The System Flowchart*



Methodology — *The System Flowchart*



Conclusion

Key Takeaways

Organized Living:

CampusMate proves that students do not need complicated, bloated websites to keep their academic lives organized.

What We Learned:

Building this project taught us how to design software with real-world users in mind, solve everyday frustrations, and manage databases safely.

Simple is Smart:

By using basic C++ tools and local text files, we built a tool that is lightning-fast, private, and works 100% offline.

Next Steps:

We strive to broaden our system by incorporating additional coding types to improve usability, enabling students to easily access everything they need while on the move.

CampusMate

A STUDENT ASSISTANT SYSTEM

Thank You

BY GROUP 14

Semester Project on *Algorithm Design and Implementation Using C++*